Exercise 2.9.11. Explain why a representation of a Lie algebra is the same thing as a representation of its universal enveloping algebra.

Example 2.9.12. The associative algebra $\mathcal{U}(\mathfrak{sl}(2))$ is the algebra generated by e, f, h, with relations

$$he - eh = 2e, \qquad hf - fh = -2f, \qquad ef - fe = h.$$

Example 2.9.13. The algebra $\mathcal{U}(\mathcal{H})$, where $\mathcal{H}$ is the Heisenberg Lie algebra, is the algebra generated by x, y, c with the relations

$$yx - xy = c, \qquad yc - cy = 0, \qquad xc - cx = 0.$$

Note that the Weyl algebra is the quotient of $\mathcal{U}(\mathcal{H})$ by the relation $c = 1$.

Remark 2.9.14. Lie algebras were introduced by Sophus Lie (see Section 2.10) as an infinitesimal version of **Lie groups** (in early texts they were called "infinitesimal groups" and were called Lie algebras by Hermann Weyl in honor of Lie). A Lie group is a group G which is also a manifold (i.e., a topological space which locally looks like $\mathbb{R}^n$) such that the multiplication operation is differentiable. In this case, one can define the algebra of smooth functions $C^\infty(G)$ which carries an action of G by right translations ($(g \circ f)(x) := f(xg)$), and the Lie algebra $\mathrm{Lie}(G)$ of G consists of derivations of this algebra which are invariant under this action (with the Lie bracket being the usual commutator of derivations). Clearly, such a derivation is determined by its action at the unit element $e \in G$, so $\mathrm{Lie}(G)$ can be identified as a vector space with the tangent space T_eG to G at e.

Sophus Lie showed that the attachment $G \mapsto \mathrm{Lie}(G)$ is a bijection between isomorphism classes of simply connected Lie groups (i.e., connected Lie groups on which every loop contracts to a point) and finite dimensional Lie algebras over $\mathbb{R}$. This allows one to study (differentiable) representations of Lie groups by studying representations of their Lie algebras, which is easier since Lie algebras are "linear" objects while Lie groups are "nonlinear". Namely, a finite dimensional representation of G can be differentiated at e to yield a representation of $\mathrm{Lie}(G)$, and conversely, a finite dimensional representation of $\mathrm{Lie}(G)$ can be exponentiated to give a representation of G. Moreover, this correspondence extends to certain classes of infinite dimensional representations.